

Table 1: Cluster Validation Metrics (External)

Metric	Formula & Variables	Role & Example
Adjusted Rand Index (ARI)	$ARI = \frac{RI - Expected_{RI}}{MaxRI - Expected_{RI}}$ RI: Rand Index (pair agreements between clustering and ground truth) Expected _{RI} : value expected by chance Max RI: maximum Rand Index	Measures similarity between predicted clustering and true labels. Example: ARI = 1 meaning perfect match with actual labels.

Table 2: Cluster Validation Metrics (Internal and Relative)

Metric	Formula & Variables	Role & Example
Silhouette Coefficient	$s(i) = \frac{b(i) - a(i)}{\max[a(i), b(i)]}$, where $a(i)$: avg distance from point i to other points in same cluster $b(i)$: avg distance from point i to points in nearest other cluster	Indicates how well points fit their clusters; range -1 to 1. Example avg silhouette 0.8 means good clustering.
Elbow Method (WCSS)	$WCSS = \sum_{k=1}^K \sum_{x_i \in C_k} \ x_i - \mu_k\ ^2$, where K : number of clusters, x_i : data point, μ_k : cluster centroid	Evaluates compactness of clusters for different K ; "elbow" point guides optimal K . Example: select K at WCSS elbow.

Table 3: Outcome Description and Example for Classification

Outcome	Description & Example
True Positive (TP)	The model correctly predicts a positive case that is actually positive. Example: In cancer detection, a patient truly has cancer and the test correctly reports "cancer present".
False Positive (FP)	The model incorrectly predicts a positive case that is actually negative. Example: In spam detection, a regular email is classified as spam by the model.
True Negative (TN)	The model correctly predicts a negative case that is actually negative. Example: In disease testing, a healthy person is identified as healthy by the test.
False Negative (FN)	The model incorrectly predicts a negative case that is actually positive. Example: In disease detection, a patient with the disease is told "no disease" by the test.

Table 4: Classification Metrics (Part 1)

Metric	Formula & Variables	Role & Example
Accuracy	$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$, where TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives	Fraction of correct predictions. Example: 0.95 means 95% of predictions are correct.
Precision	$Precision = \frac{TP}{TP+FP}$	Portion of predicted positives that are true positives. Example: 0.9 means 90% predicted positives are correct.

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Table 4 – continued from previous page

Metric	Formula & Variables	Role & Example
Recall (Sensitivity)	$Recall = \frac{TP}{TP+FN}$	Portion of actual positives correctly predicted. Example: 0.85 means 85% of actual positives detected.

Table 5: Classification Metrics (Part 2)

Metric	Formula & Variables	Role & Example
F1 Score	$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$	Harmonic mean of precision and recall. Example: 0.87 balances false positives and negatives.
Specificity	$Specificity = \frac{TN}{TN+FP}$	Fraction of negatives correctly identified; Example: 0.9 means 90% negatives correct.
AUC-ROC	Prob. model ranks random positive higher than negative; ROC plots TPR vs. FPR	Measures discrimination power across thresholds; AUC 0.95 means very good class distinction.

Table 6: Regression Metrics (Part 1)

Metric	Formula & Variables	Role & Example
Mean Squared Error (MSE)	$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, n : samples, y_i : actual, $\hat{y}_i$: predicted	Avg squared error on predictions. Example: 0.1 means small squared errors.
Root Mean Squared Error (RMSE)	$RMSE = \sqrt{MSE}$	Square root of MSE, error in units of y . Example: 0.3 means avg error of 0.3 units.
Mean Absolute Error (MAE)	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $ Example: Actual=[25,15,20,30,40], Predicted=[28,14,22,29,38], MAE = $(3+1+2+1+2)/5=1.8$	Avg magnitude of error. Example: 1.8 means predictions off by 1.8 units on average.

Table 7: Regression Metrics (Part 2)

Metric	Formula & Variables	Role & Example
Coefficient of Determination (R^2)	$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	Variance explained by model. Example: 0.85 means 85% of variance explained.
Adjusted R^2	$AdjR^2 = 1 - (1 - R^2) \times \frac{n-1}{n-p-1}$, p : predictors, n : samples	Penalizes model complexity to avoid overfitting. Example: Adjusted R^2 reduces if irrelevant features added.

What is the ROC Curve?

The Receiver Operating Characteristic (ROC) curve is a graph that illustrates how well a binary classifier separates two classes (positive and negative) as you vary the decision threshold. It is especially useful for assessing the performance of models that output probabilities or continuous scores rather than hard classifications.

- True Positive Rate (TPR, sensitivity): Proportion of actual positives correctly identified, i.e.,

$$TPR = \frac{TP}{TP + FN}$$

- False Positive Rate (FPR): Proportion of actual negatives incorrectly predicted positive, i.e.,

$$FPR = \frac{FP}{FP + TN}$$

The ROC curve plots TPR (y-axis) against FPR (x-axis) for all possible threshold values.

What is AUC (Area Under Curve)?

AUC, or Area Under the ROC Curve, is a single scalar value summarizing the entire ROC curve. It represents the probability that the model will rank a randomly chosen positive example higher than a randomly chosen negative example.

- $AUC = 1.0$: Perfect discrimination (model is always correct)
- $AUC = 0.5$: No discrimination (model's predictions are random)
- $AUC < 0.5$: Model is worse than random (systematically wrong predictions)

How is the ROC Curve Constructed?

1. Obtain model scores or probabilities for each test sample.
2. Sort samples by predicted score and try every possible threshold to separate positives from negatives.
3. At each threshold, calculate TPR and FPR.
4. Plot (FPR, TPR) pairs and connect the points.

Example

Suppose you have a spam filter that outputs a probability for each email being spam.

Email	True Label	Model Score
A	Spam	0.9
B	Ham	0.8
C	Spam	0.7
D	Ham	0.55
E	Spam	0.4
F	Ham	0.3

- Start with a high threshold (say, 0.95): All emails are predicted "not spam", so $TPR=0$, $FPR=0$.
- Gradually lower the threshold, each time recording how many true/false positives/negatives you get.
- After each change (i.e., each time an email crosses the threshold), calculate new TPR and FPR; plot these points.

If all spam emails end up with higher scores than all ham, the ROC curve will reach the top-left corner and $AUC=1$.

If some hams have higher scores than spams, AUC is less than 1.

Interpreting AUC

In practice, the AUC equals the probability that a randomly chosen spam email will be scored higher than a randomly chosen ham.

- If your spam filter has $AUC=0.95$, it means there's a 95% chance that a random spam message will be ranked above a random ham message.
- If $AUC=0.5$, the classifier is no better than flipping a coin.

Why is AUC-ROC Useful?

- Threshold independent: Considers all possible thresholds.
- Comparing models: Useful to compare classifiers, especially when the class distribution is balanced.

Limitations

- Not ideal for highly imbalanced datasets — in such cases, the Precision-Recall curve may be more informative.